8-3 Communication Protocol

Point!

1 Communication Protocol

(1) ([1]**Packet**) : The unit used when data is divided into small chunks to be transmitted over a network. Sent with a ([2]**header**) that includes the destination information.

(2) ([3]**Communication protocol (protocol)**) : A common agreement in information and communication networks.

[1] ([4]**TCP**) : A protocol that divides the data to be sent into packets, arranges the received packets in order, and requests the retransmission of any packets that were lost during communication.

[2] ([5]**UDP**) : A protocol that emphasizes sending data in real time. Used for voice calls and streaming video.

[3] ([6]**IP**) : The protocol that assigns IP addresses to deliver packets to the correct destination.

2 The Mechanism of Communication in TCP/IP

(1) ([7]**TCP/IP**) : A set of protocols used on the internet. The sending and receiving of data is controlled across four layers.

<The 4-Layer Model of TCP/IP>

Layer	Layer names	Main functions	Examples of protocols
Four layers	([8]**Application layer**)	Communication between applications	HTTP, SMTP, etc.
Three layers	([9]**Transport layer**)	Communication control, error detection, and data retransmission	TCP, UDP, etc.
Two layers	([10]**Internet layer**)	IP address allocation and routing decision	IP, etc.
One layer	([11]**Network interface layer**)	Physical connection, interaction between devices	Ethernet, etc.

(2) Mechanism of Data Communication on the internet

1) Divide the data you want to send into ([12]**packets**).

2) Attach the sender or recipient ([13]**IP addresses**) to the header of each packet.

3) Select the optimal communication route for each packet (this is called ([14]**routing**)) and deliver it to the destination.

4) If there any packets are missing upon receipt, a retransmission is requested. The packets are then rearranged in order to complete the data in its entirety.

Warm Up

Answer the following questions.

(1) For the following statements **A** to **D** concerning protocols, mark "○" if the statement is appropriate, and "×" if it is incorrect.

A A communication protocol refers to enhancing security by encrypting data.

B UDP is a protocol that ensures reliable communication by guaranteeing data integrity and order.

C TCP is a protocol that prioritizes data transfer speed, and it ensures the integrity and order of data while minimizing latency.

D IP is a protocol used to assign IP addresses to deliver packets to the correct destination.

(2) Data communication on the internet is transmitted according to TCP/IP. Rearrange the following steps **A** to **D** in the order of communication on the internet.

A Assign IP address of the recipient or the sender.

B Split the data into packets.

C Select the destination and route for the data and deliver it to its destination.

D Reorder the packets in sequence.

Explanation

(1) **A** A communication protocol is a common agreement in information and communication networks. Therefore, ×

B UDP is a protocol that emphasizes sending data in real-time. To prioritize data transfer speed, it does not guarantee the integrity or order of the data. Therefore, ×

C TCP is a protocol that divides the data to be sent into packets and retransmits any packets that are lost during transmission, ensuring reliable communication. Therefore, ×

D ○

(2) **B → A → C → D**

Try

1 Choose the most appropriate explanation of a communication protocol from the options **A** to **D** below, and answer using the letter.

A Cable hub used for connecting devices such as computers and printers to a LAN
B Description of the location of information specified in a web browser and how to retrieve it
C The number used to identify a computer in internet communication
D Protocols for communication over a network

2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [7] in the following sentences from the options **A** to **H** below, and answer using the letters.

> The rules and procedures for communication over an information and communication network are called ([1]). On the internet, ([2]) is primarily used. Among them, ([3]) is responsible for dividing the data to be sent into ([4]), rearranging the received ([4]) in order, and retransmitting any data that may be missing during communication. On the other hand, ([5]) is responsible for attaching the ([6]), which is the destination address, to ([4]) and sending them to the destination. At this time, ([4]) are transmitted via the internet, searching for the most efficient route to the recipient computer. This is referred to as ([7]).

A TCP/IP **B** IP **C** TCP **D** Packets
E IP address **F** MAC address **G** Routing **H** Communication protocol

(2) The layers in TCP/IP for data communication over the internet are summarized in the following table. Choose the term that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

Layer	Layer names	Main functions
Four layers	([1])	Communication between applications
Three layers	([2])	Communication control, error detection, and data retransmission
Two layers	([3])	IP address allocation, routing decision
One layer	([4])	Physical connection, interaction between devices

A Transport layer **B** Network interface layer
C Internet layer **D** Application layer

(3) Choose the appropriate example for each of the protocols [1] to [4] from the options **A** to **D** below, and answer using the letters.

[1] Transport layer [2] Network interface layer
[3] Internet layer [4] Application layer

A UDP **B** IP **C** SMTP **D** Ethernet

Exercise

1 Choose the term that best fits into the blanks in the following sentences from the options **A** to **D** below, and answer using the letters.

(1) A unique identification number equivalent to an address assigned to devices on an information and communication network is referred to as an ( ).

(2) A common set of rules within an information and communication network is referred to as a ( ).

A IP address **B** TCP/IP **C** Protocol **D** Route

2 Answer the following questions.

(1) Choose the most closely related options for the following sentences from the options **A** to **D** below, and answer using the letters.

[1] A protocol that emphasizes sending data in real time. Used for voice calls and streaming video.

[2] Data that is divided when communicating over a network.

[3] Protocols used on the internet. Divided into four layers.

[4] A protocol that divides the data to be sent into packets and retransmits any packets that are lost during transmission.

A TCP/IP **B** TCP **C** Packet **D** UDP

(2) The layers in TCP/IP for data communication over the internet are summarized in the following table. Choose the function that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

Layer	Layer names	Main functions	Examples of protocols
Four layers	Application layer	([1])	HTTP, SMTP, etc.
Three layers	Transport layer	([2])	TCP, UDP, etc.
Two layers	Internet layer	([3])	IP, etc.
One layer	Network interface layer	([4])	Ethernet, etc.

A Physical connection, interaction between devices
B Communication between applications
C IP address allocation, routing decision
D Communication control, error detection, and data retransmission

(3) The following statements describe the mechanism of data communication on the internet. Answer the following questions.

1) Divide the data you want to send into (**A**).
2) Assign the (**B**) of the sender or recipient to the header of each packet.
3) Select the optimal communication route for each packet and deliver it to its destination.
4) If there any packets are missing upon receipt, a retransmission is requested. The packets are then rearranged in order to complete the data in its entirety.

[1] Complete the sentences by filling in the blanks **A** and **B** with the appropriate terms.

[2] Answer with the appropriate term for the underlined section.